

## Article

# The Application of Machine Learning Methods to Predict the Power Output of Internal Combustion Engines

Ruomiao Yang <sup>1</sup>, Tianfang Xie <sup>2</sup> and Zhentao Liu <sup>1,\*</sup>

<sup>1</sup> Power Machinery and Vehicular Engineering Institute, College of Energy Engineering, Zhejiang University, Hangzhou 310027, China; 12027020@zju.edu.cn

<sup>2</sup> School of Aeronautics and Astronautics, Purdue University, West Lafayette, IN 47907, USA; tianfangxie@purdue.edu

\* Correspondence: liuzt@zju.edu.cn

**Abstract:** The indicated mean effective pressure (IMEP) is a key parameter for measuring the power output of an internal combustion engine (ICE). This indicator can be used to locate the high efficiency regions of engines. Therefore, it makes sense to predict the IMEP based on the machine learning (ML) approaches. However, different ML models are applicable to different scenarios, so it is important to choose the right model for prediction. The objective of this paper was to compare three ML models' (ANN, SVR, RF) predictive performance in forecasting IMEP indicator with the input parameters spark timing (ST), speed and load. A validated one-dimensional (1D) computational fluid dynamics (CFD) model was employed to provide 756 sets of data for the training, validation, and testing of the model. The results indicated that the random forest (RF) model had the worst prediction performance, and support vector regression (SVR) had a slightly better prediction performance than the artificial neural network (ANN), at least for the investigations in this study. Overall, the ANN and SVR models showed good predictive performance for IMEP, as the coefficient of determination ( $R^2$ ) was close to unity, and the root mean squared error (RMSE) was close to zero. Whereas the overall prediction results of the RF model are acceptable, the RF model does not learn well for some internal engine laws.

**Keywords:** spark-ignition engine; machine learning; artificial neural network; support vector regression; random forest; indicated mean effective pressure



**Citation:** Yang, R.; Xie, T.; Liu, Z. The Application of Machine Learning Methods to Predict the Power Output of Internal Combustion Engines. *Energies* **2022**, *15*, 3242. <https://doi.org/10.3390/en15093242>

Academic Editors: Dimitris Drikakis, Galih Bangga and Jaroslaw Krzywanski

Received: 21 February 2022

Accepted: 22 April 2022

Published: 28 April 2022

**Publisher's Note:** MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.



**Copyright:** © 2022 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

## 1. Introduction

With the increasing environmental pollution and energy crisis in the world, the combustion performance and exhaust emissions of engines are gradually becoming the focus of research. Additionally, the use of high-efficiency engines is gradually being promoted by governments and professionals [1,2]. Moreover, the indicated mean effective pressure (IMEP) is a key parameter for locating the area of efficient engine operation at constant speed and load [3,4]. Nowadays, ML models can play an important role in different fields, such as building [5,6], medicine [7], geography [8], energy [9,10], vehicles [11], and so on. If combining the engine research and the machine learning modeling approaches, it can help to calibrate the engine, locate the efficient region, and reduce the number of experiments/three-dimensional (3D) simulations [12,13]. Moreover, there are two significant reasons for applying machine learning in internal combustion engines. (1) It can shorten the development cycle of new engines and reduce the development costs. In particular, the design and development of low-carbon or even zero-carbon internal combustion engines are now to be carried out [14,15]. (2) Intelligent internal combustion engines will accept more data, such as climate conditions, altitude, etc., with the aim of further improving engine efficiency and reducing emissions [16,17]. Intelligent internal combustion engines need to consider data fusion, so the virtual sensor project is born, and then machine learning is needed for online updates to implement the virtual sensor [18,19]. Unlike interpolation

methods, machine learning algorithms can handle a limited amount of low-noise data and learn the intrinsic patterns of relevant parameter variations [20,21]. If the interpolation method is used to fit data with uncertainty, the generalization ability of the trained model will be poor and will not learn the intrinsic laws well [22]. Additionally, the coefficients of determination ( $R^2$ ) of the training and validation datasets will be relatively different, which indicates that the interpolation method will produce overfitting results on the data [23,24]. Moreover, although all operating conditions in the experiments are set to be the same, they are inevitably affected by ambient temperature, humidity, etc. [25].

Another advantage of machine learning is that it can fit the combustion behavior of an engine by varying the input parameters (speed, load, ST) to predict the relevant engine parameters [3,26]. Thus, it can be used as a surrogate model with sensitivity analysis and global optimization to facilitate finding the best operating point for engine control [27,28]. Moreover, many researchers have applied ML models to predict engine-related parameters. A more detailed description of the literature can be found in Table 1, and it can be found that many people have used three machine learning models (ANN, RF, SVR) to predict engine parameters [29,30]. The results show that all these ML models have good prediction of the results and are able to fit well the nonlinear relationship between the engine related output parameters and the input parameters. However, most of the studies investigated only one ML model to predict the engine-related parameters. For example, only the SVR model was used to model engine performance and emissions in ref. [20], lacking a comparison with different ML models' predicted results. Additionally, the applicability of different models under different operating conditions is different, so it is very meaningful to conduct a comparative analysis of the prediction results of different ML models [31,32]. In the field of ICE combustion, there was limited literature on comparing the predictive effects of different ML models for engine combustion. Considering that some practitioners in the energy field also lacked some knowledge about algorithm selection, the purpose of this paper is to tell which algorithm is more suitable for engine combustion prediction [33,34]. Moreover, the advantage of this study is that in addition to analyzing the model prediction results from a statistical point of view, the prediction effect is also evaluated from the perspective of ICE combustion. If this selective problem is solved, it could save a lot of time for ICE developers and guide them to design.

**Table 1.** Various ML models for engine related metrics prediction.

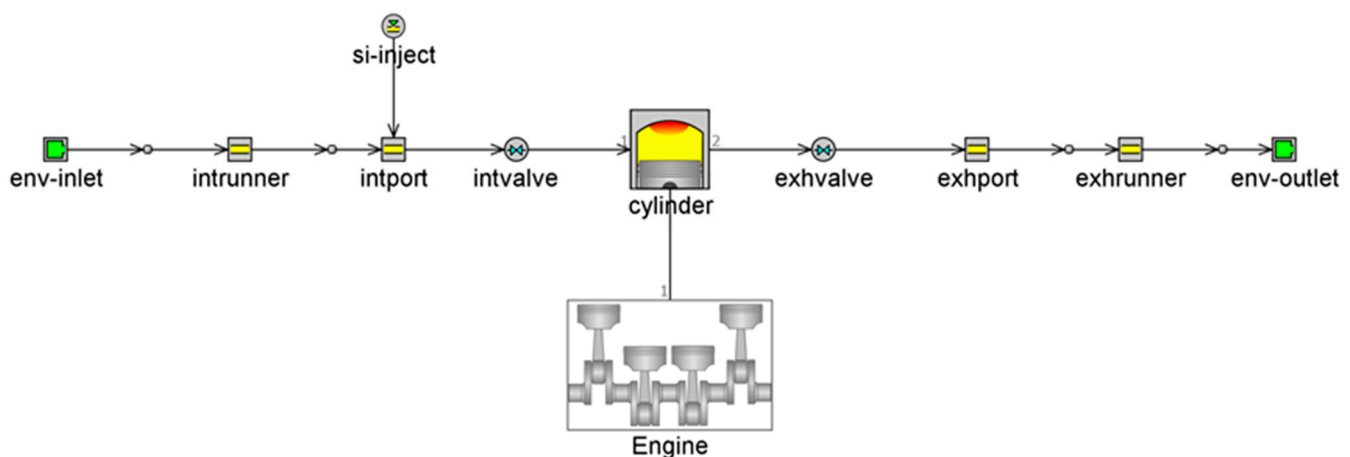
| Ref. | Model(s)           | Model's Inputs                                | Model's Output(s)               | Main Conclusions                                                                                                  |
|------|--------------------|-----------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| [20] | SVR                | Engine speed and load                         | ISFC and emissions              | The results showed that the prediction of engine performance and emissions using the SVR method is very effective |
| [35] | ANN                | ST, equivalence ratio, speed                  | PCP, PRR <sub>max</sub>         | The ANN model could be used to estimate the pressure parameters with acceptable accuracy.                         |
| [36] | ANN                | Other fuel properties                         | Fuel lubricity                  | The proposed neural network predicted the unknown data with small error.                                          |
| [37] | SVR                | Spark advance, air/fuel ratio and speed       | EGT                             | The results indicated SVR can forecast the exhaust gas temperature with acceptable errors                         |
| [38] | RF                 | Spark timing, equivalence ratio, engine speed | Combustion profile parameters   | The machine learning method presented the potential to predict the combustion behavior inside the cylinder        |
| [39] | RF                 | Spark advance, fuel/air ratio, speed          | Combustion feedback information | The work proved that the black-box approach had the potential to assist engine calibration and development        |
| [40] | RBF neural network | Time                                          | Engine system reliability       | Using ML models to predict engine failure and reliability was promising                                           |
| [41] | ANN                | Speed and load                                | Nitrogen oxides                 | The ANN model-based emission prediction results were in high agreement with the experimental values.              |
| [42] | SVR                | ST, mixture equivalence ratios, and speed     | Dynamic performance             | The established ML model could replace the more complex and time-consuming CFD model with acceptable errors.      |

In this work, a calibrated 1D CFD model will be used to provide sufficient training, validation and testing datasets. Additionally, three ML models including artificial neural network (ANN), support vector regression (SVR) and random forest (RF) will be developed to predict the indicated mean effective pressure (IMEP), which can be used to predict the efficient region. The results will compare the forecasted performance based on ML models, which can be a reference for selecting a suitable ML model to predict engine combustion-related parameters.

The rest of the paper is structured as follows: Section 2 contains a description of the numerical model, the ML models, the division of the dataset, and the model training process. Section 3 shows the prediction results of the three different algorithms. Section 4 summarizes the work of this paper and lists the main conclusions of this paper.

## 2. Materials and Methods

In this study, a four-stroke spark ignition (SI) port fuel injection (PFI) engine was used, and a validated one-dimensional computational fluid model (CFD) was utilized to provide sufficient training, validation and testing data as shown in Figure 1. The engine had a compression ratio of 9.5, bore and stroke lengths of 86 mm and 86.07 mm, respectively, and a connecting rod length of 175 mm. More detailed engine specifications can be found in Table 2, and detailed calibration information is provided in ref. [43]. The spark ignition (SI) engine was tested under different operating conditions by varying engine speed, load, and spark timings (ST) to provide sufficient data for ML models training, validation and testing. Specifically, the engine speed was varied from 1000 rpm to 6000 rpm with a 1000 rpm interval, which includes low speed to high speed. The load was adjusted by the intake pressure, which was varied from 0.5 to 1 bar in 0.1 bar increments. For the spark timing selection, in order to make the selected ST range including maximum brake torque (MBT), the range was taken to be relatively large, ranging from  $-40$  to  $0$  crank angle (CA) after top dead center (ATDC), with a spacing of 2 crank angle degrees (CAD).



**Figure 1.** A 1D CFD model of a spark-ignition engine.

**Table 2.** The parameters of the research engine.

| Research Type                  | Single-Cylinder          |
|--------------------------------|--------------------------|
| Cycle                          | 4-stroke SI PFI          |
| Valves per cylinder            | 2                        |
| Bore (mm) $\times$ Stroke (mm) | $86 \times 86.07$        |
| Intake valve opens             | 9 CAD BTDC Exhaust       |
| Intake valve closes            | 96 CAD BTDC Compression  |
| Exhaust valve opens            | 125 CAD ATDC Compression |
| Exhaust valve closes           | 38 CAD ATDC Exhaust      |
| Connecting rod length (mm)     | 175                      |
| Compression ratio              | 9.5                      |

The in-cylinder combustion and performance of internal combustion engines are affected by lots of factors, including temperature, humidity, speed, intake pressure, spark timing and other operating parameters [44,45]. Therefore, the model of an engine is a high-dimensional and nonlinear problem [46,47]. In this study, three ML models (ANN, SVR, RF) with good nonlinear fitting ability are trained to compare their predicted performance in IMEP. The three ML models are introduced as follows:

An artificial neural network (ANN) is a machine learning model that mimics the structure of a biological neural network, consisting of input, hidden, and output layers. Additionally, ANN models are composed of a number of simple, parallel and interconnected computational units connected in a specific way [48,49]. The neural network used in this study is the BP (back-propagation) algorithm, and its learning process is divided into two main phases, namely the forward propagation phase and the back-propagation phase. The forward propagation phase means that the input parameters are processed layer by layer in the network and then passed through the output layer by layer. The difference between the actual output result and the target desired result will be calculated according to a specific loss function. The error is calculated based on a specific loss function and enters the back-propagation phase. In the back-propagation process, the errors obtained from the forward propagation calculation are distributed layer by layer. The ANN model has many advantages: for example, it can solve complex nonlinear problems, has strong robustness and fault tolerance to noisy dataset, and performs better the larger the amount of data. However, the main disadvantage is that it requires initialization as well as training of a large number of parameters, such as network structure, weights, and thresholds, indicating that it requires heavy tuning.

Support vector regression (SVR) is a machine learning algorithm based on the theory of VC (Vapnik–Chervonenkis) dimensionality and structural risk minimization, etc. The idea is to find a hyperplane that divides the samples optimally, and the optimal division requires that the hyperplane maximizes the sample interval. Moreover, the proposed SVR method is used to solve linear problems at the beginning. Additionally, by changing the kernel function, the method can also solve high-dimensional nonlinear problems. Since SVR has a solid theoretical foundation, the model is highly interpretable. However, the prediction performance is sensitive to missing data and noise.

Random forest (RF) is a combinatorial algorithm whose basic composition is a certain number of decision trees. The random forest algorithm has the advantage of being resistant to interference and balancing errors in its application, and it has better classification results for data with missing features and unbalanced datasets [50]. To construct a random forest, it is first necessary to understand the basic principles of decision trees. A decision tree is a typical classifier that is often used in various classification problems. Literally, the overall structure of a decision tree resembles a tree, consisting of a “root”, “branches” and “leaves”, as well as nodes and edges with directions. The nodes are: leaf nodes and non-leaf nodes. Non-leaf nodes are generally used to put decision bases, while leaf nodes are used to put decision results. Additionally, the RF model is easy to implement and has low computational cost. However, it has been shown to overfit on some noisy classification or regression problems.

The 756 sets of data were divided into two groups to assess the performance of ML models. Eighty percent (605/756) of the dataset was randomly selected as the training dataset, and the remaining 20% of the steady-state points were regarded as validation dataset, which could validate the ML model’s performance with unknown data.

To further evaluate the learning effect of machine learning models on the intrinsic laws of spark ignition (SI) engines, 225 steady-state points were selected as the testing dataset. Among these points, 84% (189/225) operating points were used to test the learning performance of the relationship between engine performance and ST. In total, 8% (18/225) of datasets were used to test whether the ML models (ANN, SVR, RF) can forecast the relationship between load and IMEP. Additionally, the remaining 8% dataset (18/225) were

applied to test the learning effect of the ML models on the nonlinear relationship between engine speed and IMEP. The more detailed data distribution is shown in Figure 2.

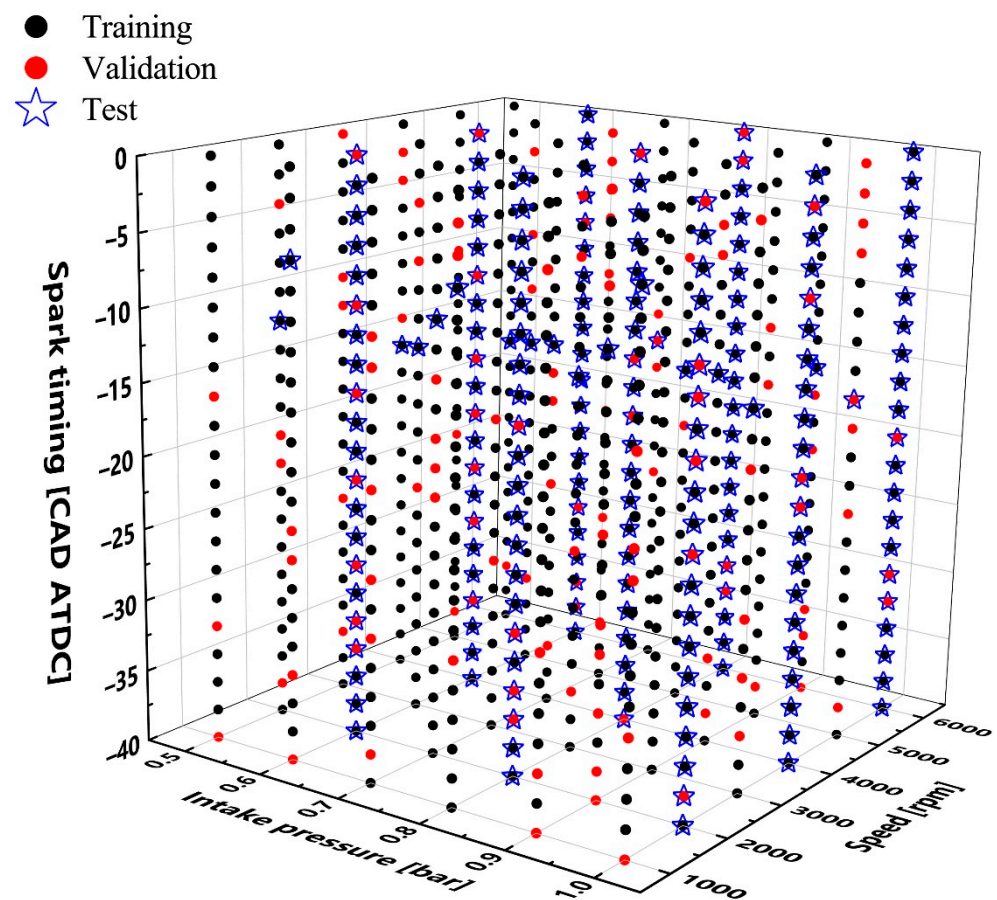
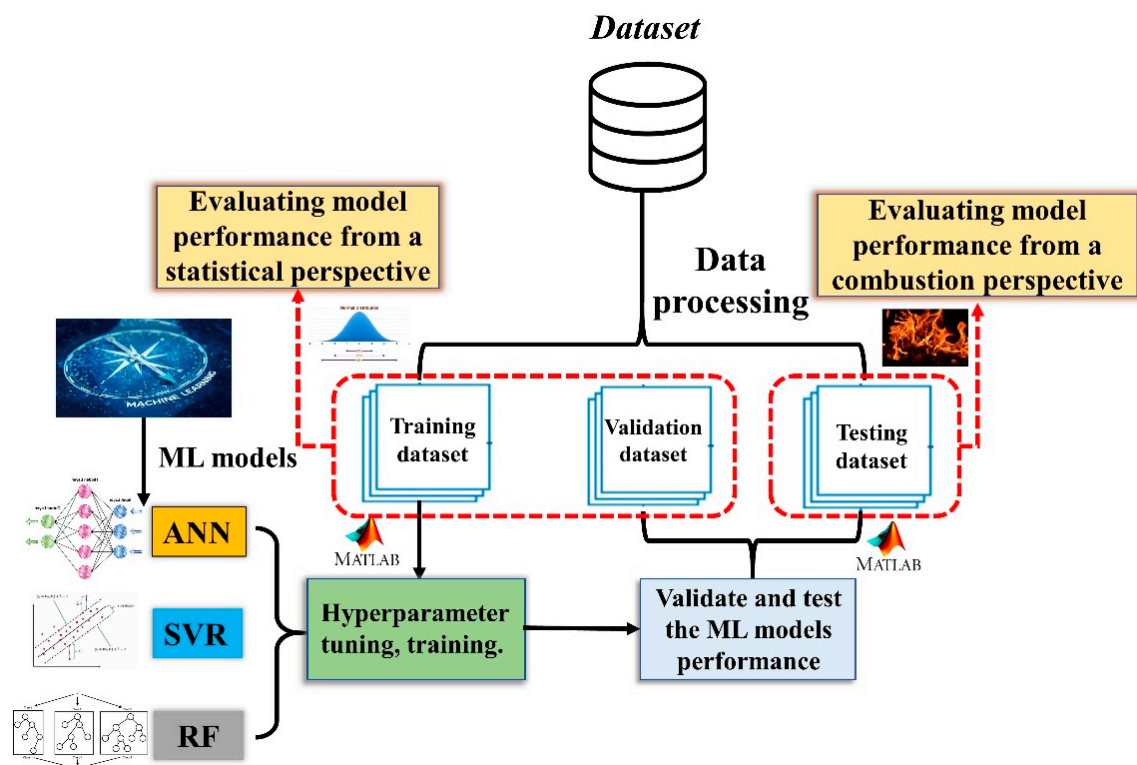


Figure 2. Training, validation and test data for ML learning algorithms.

Two key statistical indicators, including the coefficient of determination ( $R^2$ ) and the root mean square error (RMSE), were used to compare the three ML models' predicted performance. Additionally, the  $R^2$  is dimensionless, and the unit of RMSE is bar. In general, a close to one  $R^2$  and a sufficiently small RMSE represent a good prediction performance of the ML model. The detailed definition of  $R^2$  and RMSE can be found in ref. [20].

The process of applying these three ML models in predicting IMEP is shown in Figure 3. Additionally, the steps and procedures, and training and validation of the ML models are similar. At first, the initial values of the hyperparameters of the machine learning models are given, such as the number of neurons and layers of the neural network, the penalty factor and kernel function coefficients of the SVR, and the number of tree nodes and leaves of the RF model. The hyperparameters of these models are then trained through the training dataset to determine the appropriate hyperparameter values, which, in turn, builds up the entire machine learning model. Then, the validation dataset is used to judge the prediction accuracy and effectiveness of the model. Finally, the testing dataset is used to see if small errors have an effect on the intrinsic pattern of machine learning model learning.

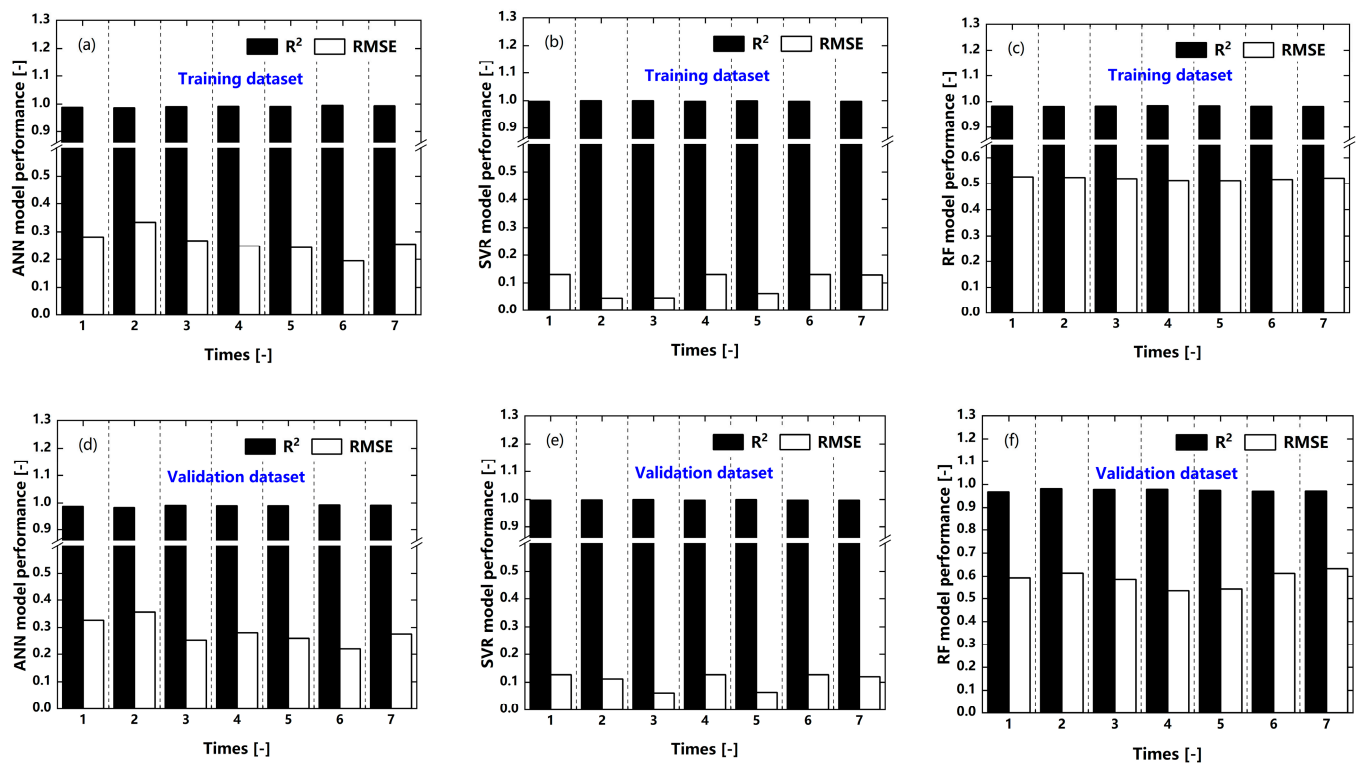


**Figure 3.** The application process of three ML models for predicting the engine-out parameter.

### 3. Results and Discussions

This section compares the robustness and performance of artificial neural network (ANN), support vector regression (SVR) and random forest (RF) on the indicated mean effective pressure (IMEP) prediction results.

To evaluate and compare the robustness of the three ML models, the three ML models in this work were trained and validated many times, and the results of seven representative ones were taken out for analysis. Figure 4 presents the  $R^2$  and RMSE values of ANN, SVR and RF models after training and validation. The results show that all three ML models have good prediction performance, as  $R^2$  is close to 1 and RMSE is relatively small for both training and validation datasets. Comparing the results of the training and validation datasets, it can be found that the  $R^2$  of the validation and training datasets are basically similar, which indicates that the model does not overfit the prediction of IMEP. Moreover, the  $R^2$  of the SVR model is the highest and the RMSE is the smallest, and the ANN is the second and the RF is the worst for the datasets studied in this paper. Moreover, for the training dataset, comparing  $R^2$  and RMSE at different running times, the prediction results of ANN and SVR models were with a certain fluctuation range, whereas the fluctuation range of the prediction results of RF model was smaller. For the validation dataset, the statistical metrics of ANN and SVR varied with the training times and remained basically consistent with those of the training dataset. On the other hand, for the validation dataset, the trend of the statistical metrics of the RF model with the training times was somewhat random and had little to do with the  $R^2$  and RMSE of the results obtained from the training dataset. The following discussion is based on the results of RUN7.

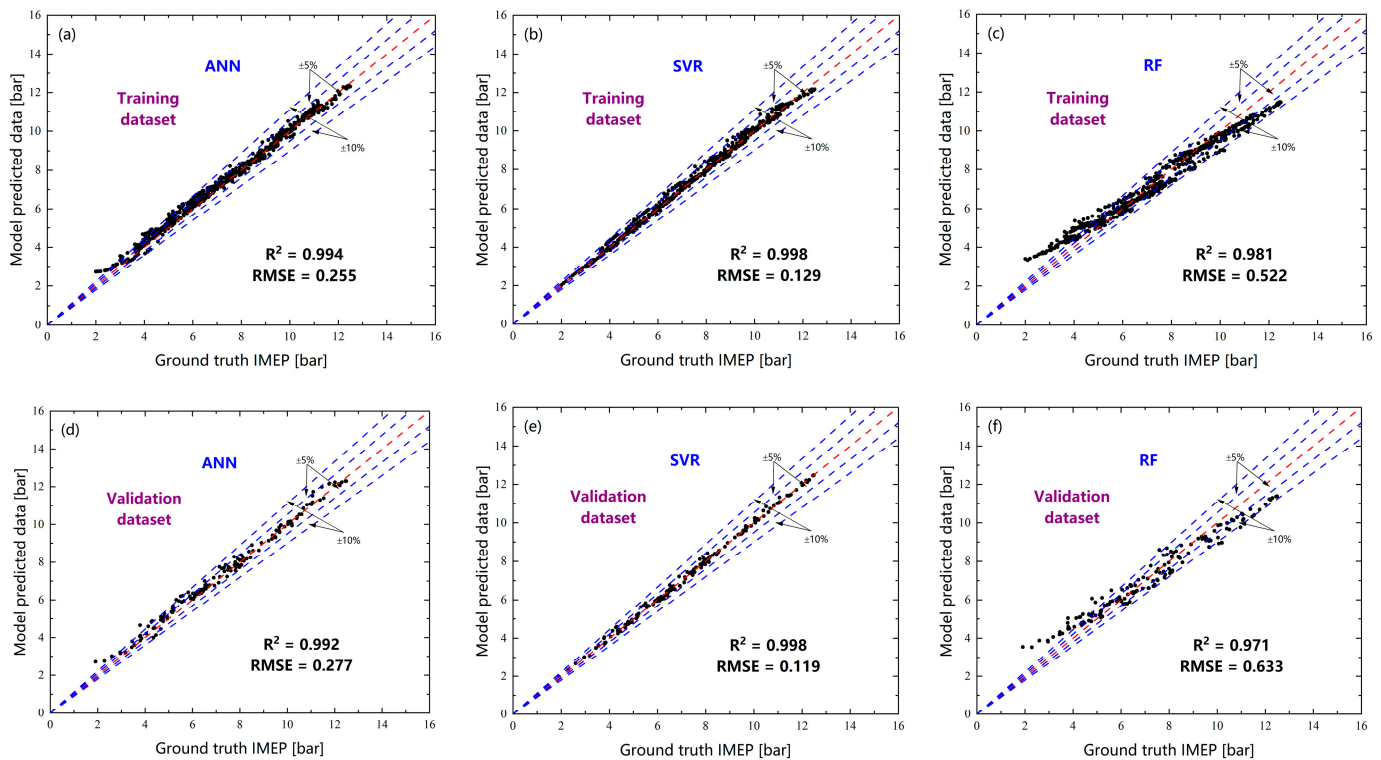


**Figure 4.** Robustness analysis of three ML models: (a) ANN model performance for the training dataset; (b) SVR model performance for the training dataset; (c) RF model performance for the training dataset; (d) ANN model performance for the validation dataset; (e) SVR model performance for the validation dataset; (f) RF model performance for the validation dataset.

Figure 5 shows the comparisons between the actual values and forecasted results based on the three ML models (ANN, SVR, RF). Figure 5a–c present the results for the 605 sets of training data, which could be used to assess the effect of ML model training. In detail, the predicted and actual IMEP values were plotted on the X and Y axes. The close-to-unity  $R^2$  and small RMSE indicated that the predicted results have good agreement with the actual data. In addition, error lines of  $\pm 5\%$  and  $\pm 10\%$  are included in the plot in order to facilitate comparison of the prediction results of these three models. It could be found that the points of the SVR model were closest to the diagonal line, followed by ANN, and the RF model had the largest deviation. Moreover, both ANN and SVR had high  $R^2$ , which was greater than 99%. Additionally, both ANN and SVR had smaller RMSE than RF model, which, to some extent, indicated that the ANN and SVR models have higher prediction accuracy after training. Overall, all three ML models had high  $R^2$ , indicating that they had all learned the intrinsic link between engine performance and input parameters (engine speed, load, ST), and that SVR and ANN performed better than the RF model.

Figure 5d–f show the comparison of predicted IMEP with actual values for the validation dataset based on three different ML models. For the same ML model, the comparison of statistical metrics between the validation and the training datasets reveals little change in  $R^2$  and RMSE, which indicates that the model did not overfit the data and had good prediction for the unseen data. Furthermore, for the validation dataset, the  $R^2$  values of ANN, SVR and RF models are 0.992, 0.998 and 0.971, respectively. The coefficients of determination ( $R^2$ ) of both ANN and SVR models are greater than 0.99, which indicates that the prediction results of ANN and SVR are very accurate. More importantly, for the validation dataset, the prediction error of SVR is basically within 5%, and the prediction error of the ANN model is basically within 10%, with individual points having an error greater than 10%. Then, compared to ANN and SVR, the RF model has more points where the prediction error is greater than 10%, especially concentrated in the low IMEP region. In

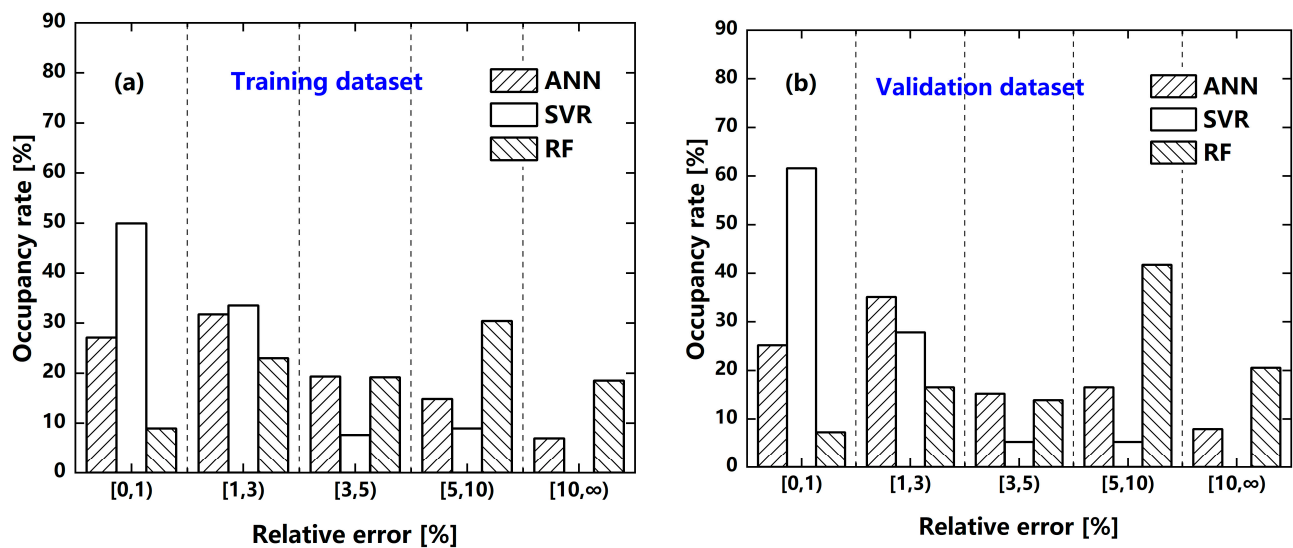
general, the prediction of SVR matches the actual value the best, followed by ANN, and RF prediction has a larger error compared to ANN and SVR.



**Figure 5.** Comparison of predicted IMEP with actual values based on different ML models: (a) ANN predicted performance for the training dataset; (b) SVR predicted performance for the training dataset; (c) RF predicted performance for the training dataset; (d) ANN predicted performance for the validation dataset; (e) SVR predicted performance for the validation dataset; (f) RF predicted performance for the validation dataset.

Figure 6 indicates the histograms of each relative error share of the prediction results for the three machine learning models with the training and validation datasets. It could be found that the SVR model predicted more than half of the data points with relative errors less than 1%, while ANN had roughly 25% of the data points with relative errors less than 1%, and the RF model had a small percentage of small error (0–1%) prediction points. For the points with large errors (>10%), the SVR had no points with errors greater than 10%, while the RF model had the largest percentage of predicted points. Therefore, the SVR predicted results are most consistent with the actual values, the ANN model has good prediction results with relative errors less than 10% for more than 90% of the points and the RF model has the worst prediction results.

The previous three figures analyzed the robustness and prediction performance of three machine learning models (ANN, SVR, RF), which indicate that both SVR and ANN outperformed the RF model. It is also of interest to assess whether small errors in prediction affect the learning effect of the machine learning models on the intrinsic laws of the internal combustion engine under various operating conditions. The pattern of influence on IMEP for a given input parameter is not very clear due to the difference in throttle position, but in general, it will follow the general law inherent to gasoline engines.

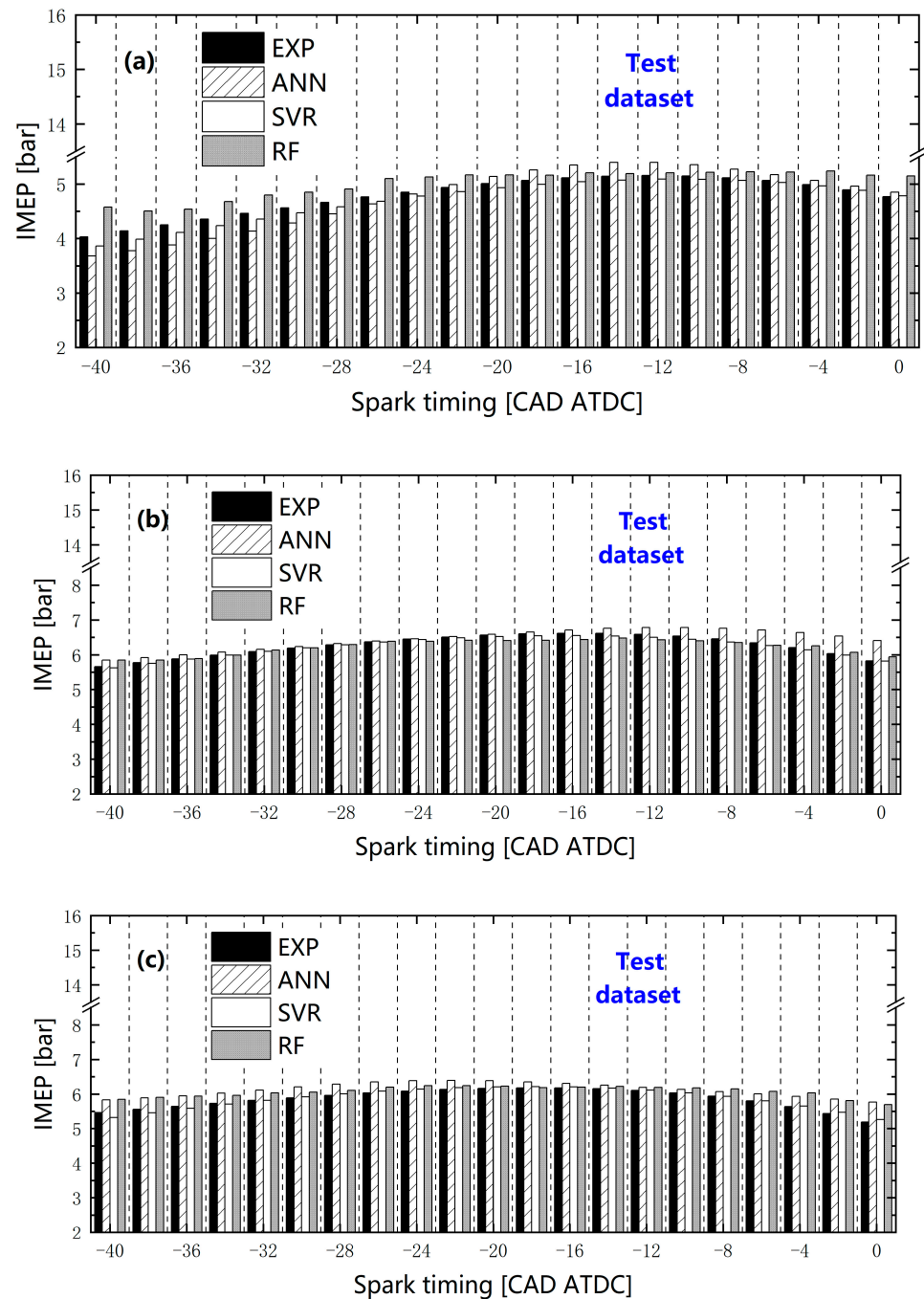


**Figure 6.** Occupancy rate in different error selection ranges based on three kinds of ML models: (a) training dataset; (b) validation dataset.

Figure 7 shows the effect of different engine speeds on the indicated mean effective pressure (IMEP) for ST at low load, where low load refers to the case where the intake pressure is 0.6 bar and low, medium and high speeds refer to 2000, 4000 and 6000 rpm, respectively. It could be noticed that the increased engine speed would delay MBT, which was in line with the general rule [51]. In addition, as the spark timing is retarded from  $-40$  CA ATDC to 0 CA ATDC, it is reflected in the actual data that IMEP first increases and then decreases due to the presence of MBT. Most importantly, the ANN and SVR models reproduced this trend, indicating that these ML models can learn the intrinsic relationship. However, as in Figure 7a, the pattern predicted by the RF model does not follow the trend of increasing and then decreasing. In addition, it could be found that the difference between the actual values and models' results is more or less 0.5 bar. However, the goal of the figure was to determine if the machine learning models can learn the intrinsic laws of an internal combustion engine. Moreover, a too-small error can lead to overfitting, and the purpose of the testing dataset is to see if this level of error is acceptable. The criterion for acceptability is whether it affects the single parameter analysis of IMEP. The purpose of predicting IMEP is to perform engine calibration. This error is acceptable as long as it does not affect the calibration process. Therefore, this level of error does not affect the learning of the engine intrinsic laws for the ML models and thus does not affect the engine calibration, so this error is acceptable. Additionally, both SVR and ANN models can learn the intrinsic law of the ST effects on IMEP with good prediction accuracy. The RF model, on the other hand, has lower prediction accuracy and does not learn the intrinsic law of the engine.

Figure 8 shows the actual values and predicted IMEP for spark timing effects based on ANN, SVR, and RF models at medium load from low to medium to high speeds. Similar to the results for low load, it can be observed from the experimental values that the IMEP first rises and then falls as the spark timing is delayed. This occurs when the spark timing is too late: for example, when the piston reaches the top dead center, the mixture begins to burn and the piston has started to move down, which will increase the cylinder volume and reduce the combustion pressure, resulting in a decrease in engine output. If the spark timing is too early, the piston is still moving to the top dead center and the cylinder pressure has reached a larger value, when the direction of gas pressure is opposite to the direction of piston movement. Moreover, the indicated effective work is reduced and the engine power decreases. Therefore, there is a suitable MBT value in order to maximize the IMEP [51]. Furthermore, it can be found the SVR prediction has the smallest error, being closer to the ground truth values in all operating conditions, while the RF model has the largest error. In

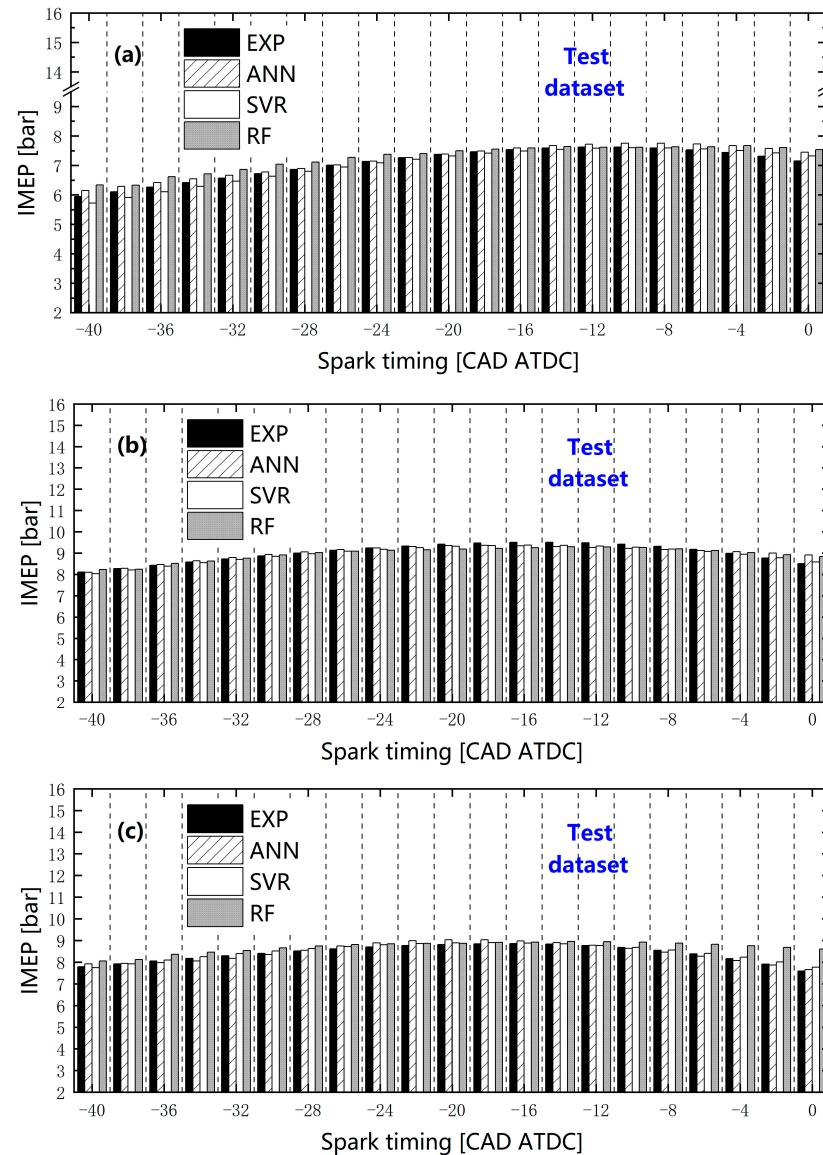
general, the trend of IMEP variation with ST is basically described by all three ML models, despite the RF model having larger relative errors.



**Figure 7.** Comparison of predicted IMEP with actual values at low load: (a) low speed; (b) medium speed; (c) high speed.

Figure 9 indicates the actual values and predicted performance metric of three ML models for ST effects under high load from low-to-medium-to-high speeds. The IMEP peaks when the ST reaches a certain spark timing, which is in accordance with the law [51]. Moreover, it can be found that the predicted values of RF model at any ST under high load are lower than the ground truth values and have larger errors. Compared to the three ML models, the prediction accuracy of SVR is the highest, followed by ANN, and the prediction error of RF model is large. For predicting the trend of IMEP, the ANN and SVR predictions

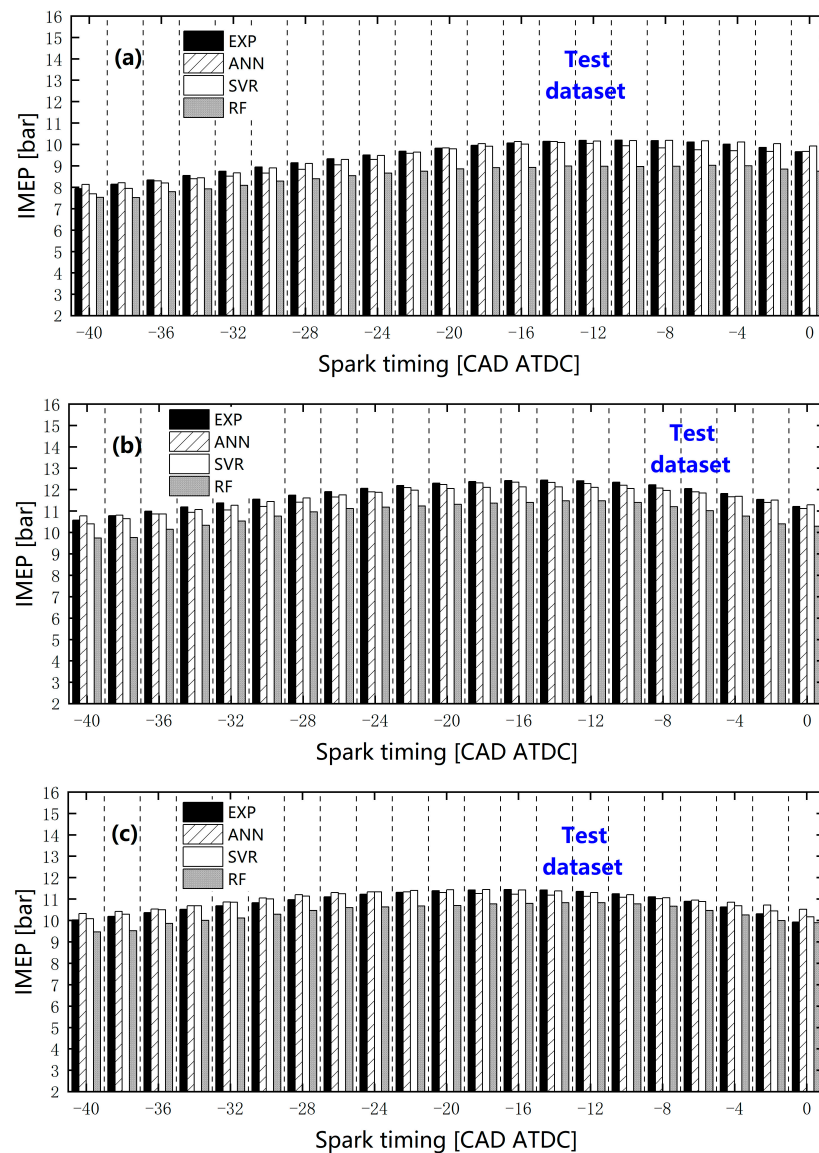
are relatively accurate, while the prediction trend of RF model was not accurate enough. Overall, for high loads, the accuracy and trend of SVR and ANN prediction results are good, and the prediction results of RF are acceptable.



**Figure 8.** Comparison of predicted IMEP with actual values at medium load: (a) low speed; (b) medium speed; (c) high speed.

Figure 10 shows the comparison of the predicted IMEP values based on the SVR, ANN, RF models with the actual values at MBT under low-to-medium-to-high loads. The graph was used to assess whether the three models could learn the law of the effect of engine speed on IMEP. As the speed increases from 1000 rpm to 4000 rpm, the IMEP increases first and then decreases in the interval from 4000 rpm to 6000 rpm. This is because, when the engine speed is low, the turbulence intensity in the cylinder is weakened, and the turbulent flame speed is smaller despite the timescale for each crank angle increases, while the heat loss increases, resulting in a lower IMEP. When the speed becomes higher, the crankshaft angle occupied by combustion increases, the constant volume becomes smaller, and sometimes, the combustion does not proceed sufficiently, resulting in a decline in IMEP. The law is lined with the general engine rule [51]. Moreover, it can be found that the prediction values of both SVR and ANN are close to the ground truth data, and the prediction error of RF model is the largest, indicating that SVR and ANN models have

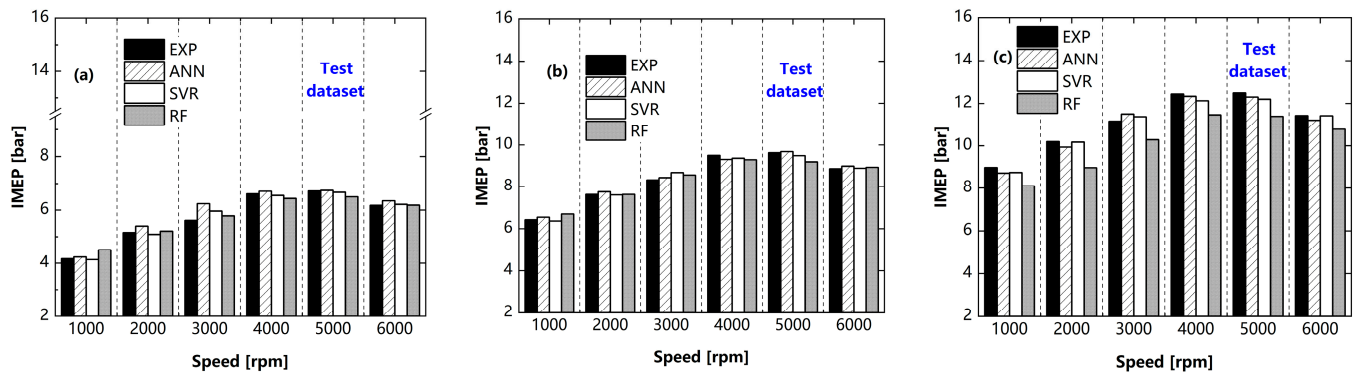
better prediction. However, the prediction laws of all three ML models are consistent with the speed characteristics of gasoline engines, showing an increasing trend followed by a decreasing trend, although the error in the prediction of the RF model is large. In general, all three models can predict the variation pattern of IMEP with speed under different loads, but SVR and ANN have better prediction accuracy than the RF model.



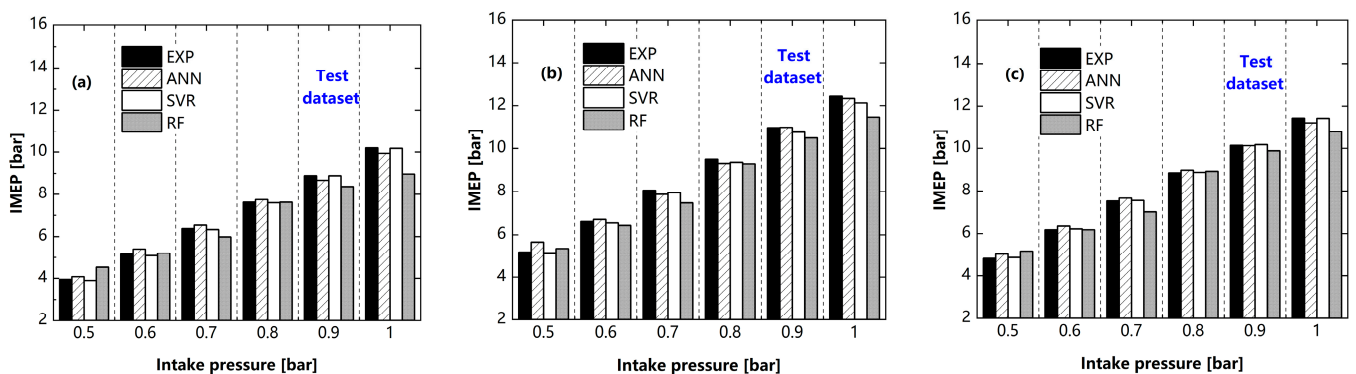
**Figure 9.** Comparison of predicted IMEP with actual values at high load: (a) low speed; (b) medium speed; (c) high speed.

Figure 11 shows the comparison of the actual values with the predicted engine performance based on different machine learning algorithms for different loads and MBT conditions. As can be seen from Figure 11, the indicated mean effective pressure gradually increases as the load increases, which is consistent with engine law [51]. This is because, as the throttle opening increases, the residual exhaust gas coefficient decreases due to the increase in in-cylinder cycle air intake. As a result, the combustion rate becomes faster, and the excess air coefficient becomes larger, resulting in better combustion and less heat transfer per unit mass. These combined factors lead to an increase in IMEP when the load becomes larger. In addition, it can be found that the white bars represented by the SVR are closest to the black bars represented by the ground truth values, which indicates that the accuracy of the SVR prediction is good. Moreover, all three machine learning models

learned the pattern of load effects on IMEP, although the RF model had a larger prediction error compared to the other two ML models. Overall, the predicted results of SVR and ANN are consistent with the actual values, while the prediction error of RF model is larger compared with ANN and SVR.



**Figure 10.** Comparison of predicted IMEP with actual values at MBT: (a) low load; (b) medium load; (c) high load.



**Figure 11.** Comparison of predicted IMEP with actual values at MBT: (a) low speed; (b) medium speed; (c) high speed.

#### 4. Conclusions

Nowadays, many studies have applied ML models to predict and optimize the engine-related parameters, and these results showed that the incorporation of machine learning algorithms in the engine domain has achieved excellent performance. However, most researchers have focused on studying one model's predicted performance, lacking different models' forecasted results comparison. Therefore, the objective of this work was to assess and compare the effectiveness of these three ML models (SVR, ANN, RF) for indicated mean effective pressure (IMEP) prediction, using spark timing, speed, and torque as input parameters. The main conclusions are as follows:

1. By comparing  $R^2$  and RMSE of different RUNs, it could be found that the robustness of all three models was good. Additionally, the statistical indicators of the training and validation datasets were close, indicating that the machine learning models did not overfit for these datasets.
2. Both ANN and SVR can learn the law of internal combustion engines well. However, the RF model cannot learn the law well, at least for the operating conditions investigated in this work.
3. For the prediction of engine related parameters, the prediction accuracy and effect of SVR and ANN were comparable. The disadvantage of ANN was that it required heavy tuning. For SVR model, it took longer time to train the algorithm. In the future, with the development of intelligent engines, less iterations are needed in the

online learning process, so it will be better to use ANN model to predict combustion-related parameters.

Overall, for the noise-free data, the prediction performance of both ANN and SVR was good, while the RF performance was poorer compared to the others. If the future development of intelligent engines and the need for online learning are considered, the ANN model is more suitable for predicting engine-related parameters. In the future, the prediction results of different machine learning models will be compared and analyzed based on noisy data.

**Author Contributions:** R.Y., Conceptualization, methodology, simulation, writing—draft preparation; T.X., Simulation, analysis; Z.L., Analysis, supervision. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research received no external funding.

**Institutional Review Board Statement:** Not applicable.

**Informed Consent Statement:** Not applicable.

**Data Availability Statement:** The data presented in this study are available on request from the corresponding author.

**Acknowledgments:** We gratefully acknowledge Zhejiang University for providing the equipment and software usage rights.

**Conflicts of Interest:** The authors declare no conflict of interest.

## Abbreviations

|                    |                                     |
|--------------------|-------------------------------------|
| 1D                 | One-dimensional                     |
| 3D                 | Three-dimensional                   |
| ANN                | Artificial neural network           |
| ATDC               | After top dead center               |
| BP                 | Back-propagation                    |
| BTDC               | Before top dead center              |
| CA                 | Crank angle                         |
| CAD                | Crank angle degree                  |
| CFD                | Computational fluid dynamics        |
| EGT                | Exhaust gas temperature             |
| ICE                | Internal combustion engine          |
| IMEP               | Indicated mean effective pressure   |
| ISFC               | Indicated specific fuel consumption |
| SVR                | Support vector regression           |
| MBT                | Maximum brake torque                |
| ML                 | Machine learning                    |
| PFI                | Port fuel injection                 |
| R <sup>2</sup>     | Coefficient of determination        |
| RBF                | Radial basis function               |
| RF                 | Random forest                       |
| RMSE               | Root mean squared error             |
| SI                 | Spark ignition                      |
| ST                 | Spark timing                        |
| PCP                | Peak cylinder pressure              |
| PRR <sub>max</sub> | Maximum pressure rise rate          |

## References

1. Liu, Z.; Zuo, Q.; Wu, G.; Li, Y. An artificial neural network developed for predicting of performance and emissions of a spark ignition engine fueled with butanol–gasoline blends. *Adv. Mech. Eng.* **2018**, *10*, 1687814017748438. [[CrossRef](#)]
2. Mohammadhassani, J.; Dadvand, A.; Khalilarya, S.; Solimanpurb, M. Prediction and reduction of diesel engine emissions using a combined ANN–ACO method. *Appl. Soft Comput.* **2015**, *34*, 139–150. [[CrossRef](#)]

3. Ma, F.K.; Wang, J.; Feng, Y.N.; Zhang, Y.G.; Su, T.X.; Zhang, Y.; Liu, Y.H. Parameter optimization on the uniflow scavenging system of an OP2S-GDI engine based on indicated mean effective pressure (IMEP). *Energies* **2017**, *10*, 368. [\[CrossRef\]](#)
4. Omran, R.; Younes, R.; Champoussin, J.C.; Outbib, R. New Indicated Mean Effective Pressure (IMEP) model for predicting crankshaft movement. *Energy Convers. Manag.* **2011**, *52*, 3376–3382. [\[CrossRef\]](#)
5. Ahmad, M.W.; Mourshed, M.; Rezgui, Y. Trees vs Neurons: Comparison between random forest and ANN for high-resolution prediction of building energy consumption. *Energy Build.* **2017**, *147*, 77–89. [\[CrossRef\]](#)
6. Sadat-Mohammadi, M.; Shakerian, S.; Liu, Y.; Asadi, S.; Jebelli, H. Non-invasive physical demand assessment using wearable respiration sensor and random forest classifier. *J. Build. Eng.* **2021**, *44*, 103279. [\[CrossRef\]](#)
7. Cai, Z.; Poulos, R.C.; Liu, J.; Zhong, Q. Machine learning for multi-omics data integration in cancer. *iScience* **2022**, *25*, 103798. [\[CrossRef\]](#)
8. Liu, Z.; Liu, J. Machine Learning Assisted Analysis of an Ammonia Engine Performance. *J. Energy Resour. Technol.* **2022**, *144*, 112307. [\[CrossRef\]](#)
9. Liu, J.; Dumitrescu, C.; Ulishney, C. Investigation of Heat Transfer Characteristics of Heavy-Duty Spark Ignition Natural Gas Engines Using Machine Learning. *SAE Tech. Pap.* **2022**. [\[CrossRef\]](#)
10. Yan, Z.; Gainey, B.; Gohn, J.; Hariharan, D.; Saputo, J.; Schmidt, C.; Caliar, F.; Sampath, S.; Lawler, B. A comprehensive experimental investigation of low-temperature combustion with thick thermal barrier coatings. *Energy* **2021**, *222*, 119954. [\[CrossRef\]](#)
11. Huang, C.; Molisch, A.F.; He, R.; Wang, R.; Tang, P.; Zhong, Z. Machine-learning-based data processing techniques for vehicle-to-vehicle channel modeling. *IEEE Commun. Mag.* **2019**, *57*, 109–115. [\[CrossRef\]](#)
12. Liu, J.; Ulishney, C.; Dumitrescu, C.E. Comparative performance of machine learning algorithms in predicting nitrogen oxides emissions of a heavy duty natural gas spark ignition engine. *Int. Conf. Appl. Energy* **2021**, *29*, ID-71.
13. Zhang, Y.; Yang, R.; Yuchao, Y.A.; Liu, Z.; Fu, J. Numerical investigations of injection timing effects on a GDI engine performance: Part B, In-cylinder emission formation and oxidation process. *Front. Energy Res.* **2022**, *10*, 823386. [\[CrossRef\]](#)
14. Liu, J.; Dumitrescu, C.E. Single and double Wiebe function combustion model for a heavy-duty diesel engine retrofitted to natural-gas spark-ignition. *Appl. Energy* **2019**, *248*, 95–103. [\[CrossRef\]](#)
15. Gainey, B.; Yan, Z.; Lawler, B. Autoignition characterization of methanol, ethanol, propanol, and butanol over a wide range of operating conditions in LTC/HCCI. *Fuel* **2021**, *287*, 119495. [\[CrossRef\]](#)
16. Yang, R.; Yan, Y.; Sijia, R.; Liu, Z.; Zhang, Y.; Fu, J. A Support Vector Machine Model to Predict a Spark Ignition Engine Performance and Emissions. *SAE Tech. Pap.* **2022**. [\[CrossRef\]](#)
17. Yan, Y.; Yang, R.; Sun, X.; Liu, Z. Numerical investigations of injection timing effects on a GDI engine performance: Part A, In-cylinder combustion process. *Front. Energy Res.* **2022**, *10*, 828167. [\[CrossRef\]](#)
18. Yang, L.; Shami, A. On hyperparameter optimization of machine learning algorithms: Theory and practice. *Neurocomputing* **2020**, *415*, 295–316. [\[CrossRef\]](#)
19. Liu, J.; Wang, H. Machine learning assisted modeling of mixing timescale for LES/PDF of high-Karlovitz turbulent premixed combustion. *Combust. Flame* **2022**, *238*, 111895. [\[CrossRef\]](#)
20. Zhang, Y.; Wang, Q.; Chen, X.; Yan, Y.; Yang, R.; Liu, Z.; Fu, J. The Prediction of Spark-Ignition Engine Performance and Emissions Based on the SVR Algorithm. *Processes* **2022**, *10*, 312. [\[CrossRef\]](#)
21. Mathew, V.; Toby, T.; Singh, V.; Rao, B.M.; Kumar, M.G. Prediction of Remaining Useful Lifetime (RUL) of turbofan engine using machine learning. In Proceedings of the 2017 IEEE International Conference on Circuits and Systems (ICCS 2017), Thiruvananthapuram, Kerala, India, 20–21 December 2017; pp. 306–311.
22. Huang, Q.; Liu, J.; Ulishney, C.; Dumitrescu, C.E. On the use of artificial neural networks to model the performance and emissions of a heavy-duty natural gas spark ignition engine. *Int. J. Engine Res.* **2021**, 14680874211034409. [\[CrossRef\]](#)
23. Gambella, C.; Ghaddar, B.; Naooum-Sawaya, J. Optimization problems for machine learning: A survey. *Eur. J. Oper. Res.* **2021**, *290*, 807–828. [\[CrossRef\]](#)
24. Lei, C.; Deng, J.; Cao, K.; Ma, L.; Xiao, Y.; Ren, L. A random forest approach for predicting coal spontaneous combustion. *Fuel* **2018**, *223*, 63–73. [\[CrossRef\]](#)
25. Liu, J.; Dumitrescu, C.E. Improved thermodynamic model for lean natural gas spark ignition in a diesel engine using a triple Wiebe function. *J. Energy Resour. Technol.* **2020**, *142*, 062303. [\[CrossRef\]](#)
26. Li, Y.; Jia, M.; Han, X.; Bai, X.S. Towards a comprehensive optimization of engine efficiency and emissions by coupling artificial neural network (ANN) with genetic algorithm (GA). *Energy* **2021**, *225*, 120331. [\[CrossRef\]](#)
27. Liu, Z.; Zhang, Y.; Fu, J.; Liu, J. Multidimensional computational fluid dynamics combustion process modelling of a 6V150 diesel engine. *J. Therm. Sci. Eng. Appl.* **2022**, *14*, 101009. [\[CrossRef\]](#)
28. Zhang, Y.; Yan, Y.; Yang, R.; Wang, Q.; Zhang, B.; Gan, Q.; Liu, Z.; Fu, J. Study on In-cylinder Heat Transfer Boundary Condition for Diesel Engine under Variable Altitudes Based on CHT Model. *Front. Energy Res.* **2022**, *10*, 828215. [\[CrossRef\]](#)
29. Aliramezani, M.; Koch, C.R.; Shahbakhti, M. Modeling, diagnostics, optimization, and control of internal combustion engines via modern machine learning techniques: A review and future directions. *Prog. Energy Combust. Sci.* **2022**, *88*, 100967. [\[CrossRef\]](#)
30. Lee, J.; Kwon, S.; Kim, H.; Keel, J.; Yoon, T. Machine Learning Applied to the NOx Prediction of Diesel Vehicle under Real Driving Cycle. *Appl. Sci.* **2021**, *11*, 3758. [\[CrossRef\]](#)

31. Heung, B.; Ho, H.C.; Zhang, J.; Knudby, A.; Bulmer, C.E.; Schmidt, M.G. An overview and comparison of machine-learning techniques for classification purposes in digital soil mapping. *Geoderma* **2016**, *265*, 62–77. [\[CrossRef\]](#)
32. Tuttle, J.F.; Blackburn, L.D.; Andersson, K.; Powell, K.M. A systematic comparison of machine learning methods for modeling of dynamic processes applied to combustion emission rate modeling. *Appl. Energy* **2021**, *292*, 116886. [\[CrossRef\]](#)
33. Poorghasemi, K.; Saray, R.K.; Bahlouli, K.; Zehni, A. 3D CFD simulation of a natural gas fueled HCCI engine with employing a reduced mechanism. *Fuel* **2016**, *182*, 816–830. [\[CrossRef\]](#)
34. Yan, Z.; Gaaney, B.; Hariharan, D.; Lawler, B. Investigation into reactivity separation between direct injected and premixed fuels in RCCI combustion mode. In Proceedings of the ASME 2017 internal combustion engine division fall technical conference, Seattle, WA, USA, 15–18 October 2017; American Society of Mechanical Engineers: New York, NY, USA, 2017; Volume 59346, p. V001T03A001.
35. Liu, J.; Ulishney, C.; Dumitrescu, C.E. Prediction of efficient operating conditions inside a heavy-duty natural gas spark ignition engine using artificial neural networks. In Proceedings of the ASME 2020 International Mechanical Engineering Congress and Exposition, Online, 16–19 November 2020; American Society of Mechanical Engineers: New York, NY, USA, 2020; Volume 84560, p. V008T08A013.
36. Korres, D.M.; Anastopoulos, G.; Lois, E.; Alexandridis, A.; Sarimveis, H.; Bafas, G. A neural network approach to the prediction of diesel fuel lubricity. *Fuel* **2002**, *81*, 1243–1250. [\[CrossRef\]](#)
37. Liu, J.; Huang, Q.; Ulishney, C.; Dumitrescu, C.E. Prediction of exhaust gas temperature of a natural gas spark ignition engine using machine learning methods. In Proceedings of the 12th International Conference on Applied Energy, Virtual, Online, Bangkok, Thailand, 29 November–2 December 2020.
38. Liu, J.; Ulishney, C.; Dumitrescu, C.E. Application of random forest machine learning models to forecast combustion profile parameters of a natural gas spark ignition engine. In Proceedings of the ASME International Mechanical Engineering Congress and Exposition, Virtual, Online, 16–19 November 2020; American Society of Mechanical Engineers: New York, NY, USA, 2020; Volume 84539, p. V006T06A003.
39. Liu, J.; Ulishney, C.; Dumitrescu, C.E. Random forest machine learning model for predicting combustion feedback information of a natural gas spark ignition engine. *J. Energy Resour. Technol.* **2021**, *143*, 012301. [\[CrossRef\]](#)
40. Xu, K.; Xie, M.; Tang, L.C.; Ho, S.L. Application of neural networks in forecasting engine systems reliability. *Appl. Soft Comput.* **2003**, *2*, 255–268. [\[CrossRef\]](#)
41. Obodeh, O.; Ajuwa, C.I. Evaluation of artificial neural network performance in predicting diesel engine NOx emissions. *Eur. J. Sci. Res.* **2009**, *33*, 642–653.
42. Liu, J.; Huang, Q.; Ulishney, C.; Dumitrescu, C.E. A Support-Vector Machine Model to Predict the Dynamic Performance of a Heavy-Duty Natural Gas Spark Ignition Engine. *SAE Tech. Pap.* **2021**. [\[CrossRef\]](#)
43. Yang, R.; Sun, X.; Liu, Z.; Zhang, Y.; Fu, J. A Numerical Analysis of the Effects of Equivalence Ratio Measurement Accuracy on the Engine Efficiency and Emissions at Varied Compression Ratios. *Processes* **2021**, *9*, 1413. [\[CrossRef\]](#)
44. Sayin, C.; Ertunc, H.M.; Hosoz, M.; Kilicaslan, I.; Canakci, M. Performance and exhaust emissions of a gasoline engine using artificial neural network. *Appl. Therm. Eng.* **2007**, *27*, 46–54. [\[CrossRef\]](#)
45. Stocchi, I.; Liu, J.; Dumitrescu, C.E.; Battistoni, M.; Grimaldi, C.N. Effect of piston crevices on the numerical simulation of a heavy-duty diesel engine retrofitted to natural-gas spark-ignition operation. *J. Energy Resour. Technol.* **2019**, *141*, 112204. [\[CrossRef\]](#)
46. Yan, Z.; Gaaney, B.; Gohn, J.; Hariharan, D.; Saputo, J.; Schmidt, C.; Caliar, F.; Sampath, S.; Lawler, B. The effects of thick thermal barrier coatings on low-temperature combustion. *SAE Int. J. Adv. Curr. Pract. Mobil.* **2020**, *2*, 1786–1799.
47. Liu, J.; Dumitrescu, C.E. Investigation of multistage combustion inside a heavy-duty natural-gas spark-ignition engine using three-dimensional computational fluid dynamics simulations and the Wiebe-function combustion model. *J. Eng. Gas Turbines Power* **2020**, *142*, 101012. [\[CrossRef\]](#)
48. Fu, J.; Yang, R.; Li, X.; Sun, X.; Li, Y.; Liu, Z.; Zhang, Y.; Sunden, B. Application of artificial neural network to forecast engine performance and emissions of a spark ignition engine. *Appl. Therm. Eng.* **2022**, *201*, 117749. [\[CrossRef\]](#)
49. Norouzi, A.; Gordon, D.; Aliramezani, M.; Koch, C.R. Machine learning-based diesel engine-Out NOx reduction using a plug-in PD-type iterative learning control. In Proceedings of the 2020 IEEE Conference on Control Technology and Applications (CCTA), Montréal, QC, Canada, 24–26 August 2020.
50. Liu, J.; Huang, Q.; Ulishney, C.; Dumitrescu, C.E. Machine learning assisted prediction of exhaust gas temperature of a heavy-duty natural gas spark ignition engine. *Appl. Energy* **2021**, *300*, 117413. [\[CrossRef\]](#)
51. Heywood, J.B. *Internal Combustion Engine Fundamentals*; McGraw-Hill: New York, NY, USA, 1988.